Aditya Jadhav

520-4756461 | adityaj2003@gmail.com | github.com | linkedin.com | Personal Website

Education

University of Arizona January 2021 - December 2024
BS Computer Science *GPA: 3.86/4.00*

Technical Skills

Languages: Java, Python, JavaScript, C, SQL
Frameworks/Tools: Docker, Kubernetes, React.js, AWS, Node.js, MongoDB, Postgres, Express.js
Libraries: TensorFlow, Apache Spark, Pandas, NumPy
Relevant Coursework: Machine Learning, Distributed Computing, Algorithms, OOP, Computer Vision, OS, Networking

Experience

Software Engineering Intern Oct 2023 – Jun 2024, Mar 2025 – Present
Astrocomm Technologies | ***Python, C, MPI, ElasticSearch, wxPython***

- Designed and developed real-time metric dashboards using **wxPython**, integrating with **Elasticsearch** for low-latency search and analytics.
- Optimized **MPI** distributed data pipelines, reducing latency and improving throughput for large-scale analytics.
- Implemented and evaluated **RNN/LSTM**-based models to predict thermal performance in SSPAs, enabling early detection of potential failure conditions.
- Used **Qucs-S** to script and simulate GaN HEMT models, supporting circuit-level validation of thermal behavior.

Undergraduate Research Assistant – Machine Learning Theory February 2023 – January 2024
University of Arizona | ***Python, Scikit-learn, NumPy***

- Researched label-efficient active learning under non-convex loss functions with Dr. Chicheng Zhang.
- Implemented and tuned **Proximal Stochastic Gradient Descent** variants under structured noise assumptions (Tsybakov, Massart), demonstrating improved convergence behavior.
- Configured **SLURM** workloads and session-managed experiments via **TMUX** on UA HPC, achieving **30%** runtime reduction for GPU training pipelines.

Undergraduate Research Assistant November 2023 – October 2024
Visual and Autonomous Exploration Lab | ***Python, Hadoop, Spark, PyTorch***

- Engineered modular deep learning architectures from scratch with custom activations and optimizers for bias analysis and robustness testing.
- Developed methods to detect and mitigate model bias, improve fairness, interpretability, and reliability.
- Leveraged **Hadoop** and **Spark** for distributed ML workloads, enabling scalablability across large datasets.

Projects

Chess Video to PGN Converter | ***PyTorch, OpenCV, AWS Lambda, S3, SageMaker***

- Built a computer vision system to convert chess game videos into PGN as a low-cost alternative to e-boards.
- Trained **PyTorch CNNs** on 500+ images for board and piece recognition using **OpenCV**-based preprocessing, achieving **90%** test accuracy.
- Developed an end-to-end pipeline with **Flask**, **AWS Lambda**, **S3**, and **SageMaker** to handle video upload, frame extraction, and prediction.

IBM Watson-Inspired Q&A Program | ***Java, Lucene, Python, BeautifulSoup4***

- Implemented a question-answering system inspired by IBM Watson, using **Apache Lucene** and **TF-IDF** to index and rank answers from Wikipedia.
- Incorporated **query expansion** using synonym detection and n-gram matching to improve retrieval relevance.
- Built a custom webcrawler in **Python** with **BeautifulSoup4** to gather and preprocess large-scale Wikipedia data.

Semantic Search PDF App | ***Swift, CoreML, PyTorch, C++***

- Built a semantic search tool for PDF documents using **MiniLM**-based embeddings to support concept-level information retrieval.
- Implemented vector indexing with **HNSW graphs** in **C++**, achieving **50% faster** indexing performance over standard implementation.
- Converted a pretrained **MiniLM** model from **PyTorch** to **CoreML** and integrated it with **parallelized Swift** routines for efficient on-device embedding generation.